

Overview of Vision transformer

Subject: Vision transformer

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

1.

The Masked Autoencoder took inspiration from denoising autoencoders and context encoders. It has two ViTs put end-to-end. The first one ("encoder") takes in image patches with positional encoding, and outputs vectors representing each patch. The second one (called "decoder", even though it is still an encoder-only Transformer) takes in vectors with positional encoding and outputs image patches again.

2.

Others Other examples include the visual transformer, CoAtNet, CvT, the data-efficient ViT (DeiT), etc. In the Transformer in Transformer architecture, each layer applies a vision Transformer layer on each image patch embedding, add back the resulting tokens to the embedding, then applies another vision Transformer layer.

3.

LayerNorm immediately after each attention and feedforward layer ("res-post-norm"); scaled cosine attention to replace the original dot product attention; log-spaced continuous relative position bias, which allows transfer learning across different window resolutions.

4.

Comparison with CNNs Typically, ViT uses patch sizes larger than standard CNN kernels (3x3 to 7x7). ViT is more sensitive to the choice of the optimizer, hyperparameters, and network depth. Preprocessing with a layer of smaller-size, overlapping (stride < size) convolutional filters helps with performance and stability. This different behavior seems to derive from the different inductive biases they possess:

5.

Prediction In prediction, the decoder architecture is discarded entirely. The input image is split into patches by the same algorithm as in training, but no patches are masked out. A linear projection with a positional embedding is applied to each patch, and the resulting embedding vector representations of each patch are fed to the encoder.

6.

After the ViT processes an image, it produces some embedding vectors. These must be converted to a single class probability prediction by some kind of network. In the original ViT and Masked Autoencoder, they used a dummy [CLS] token, in emulation of the BERT language model. The output

at [CLS] is the classification token, which is then processed by a LayerNorm-feedforward-softmax module into a probability distribution. Global average pooling (GAP) does not use the dummy token, but simply takes the average of all output tokens as the classification token. It was mentioned in the original ViT as being equally good. Multihead attention pooling (MAP) applies a multiheaded attention block to pooling. Specifically, it takes as input a list of vectors $x_1, x_2, \dots, x_n$, which might be thought of as the output vectors of a layer of a ViT. The output from MAP is $\text{MultiheadedAttention}(Q, V, V)$, where q is a trainable query vector, and V is the matrix with rows being $x_1, x_2, \dots, x_n$. This was first proposed in the Set Transformer architecture. Later papers demonstrated that GAP and MAP both perform better than BERT-like pooling. A variant of MAP was proposed as class attention, which applies MAP, then feedforward, then MAP again. Re-attention was proposed to allow training deep ViT. It changes the multiheaded attention module.

7.

History Transformers were introduced in Attention Is All You Need (2017), and have found widespread use in natural language processing. A 2019 paper applied ideas from the Transformer to computer vision. Specifically, they started with a ResNet, a standard convolutional neural network used for computer vision, and replaced all convolutional kernels by the self-attention mechanism found in a Transformer. It resulted in superior performance. However, it is not a Vision Transformer. In 2020, an encoder-only Transformer was adapted for computer vision, yielding the ViT, which reached state of the art in image classification, overcoming the previous dominance of CNN. The masked autoencoder (2022) extended ViT to work with unsupervised training. The vision transformer and the masked autoencoder, in turn, stimulated new developments in convolutional neural networks. Subsequently, there was cross-fertilization between the previous CNN approach and the ViT approach. In 2021, some important variants of the Vision Transformers were proposed. These variants are mainly intended to be more efficient, more accurate or better suited to a specific domain. Two studies improved efficiency and robustness of ViT by adding a CNN as a preprocessor. The Swin Transformer achieved state-of-the-art results on some object detection datasets such as COCO, by using convolution-like sliding windows of attention mechanism, and the pyramid process in classical computer vision.

8.

A vision transformer (ViT) is a transformer designed for computer vision. A ViT decomposes an input image into a series of patches (rather than text into tokens), serializes each patch into a vector, and maps it to a smaller dimension with a single matrix multiplication. These vector embeddings are then processed by a transformer encoder as if they were token embeddings. ViTs were designed as alternatives to convolutional neural networks (CNNs) in computer vision applications. They have different inductive biases, training stability, and data efficiency. Compared to CNNs, ViTs are less data efficient, but have higher capacity. Some of the largest modern computer vision models are ViTs, such as one with 22B parameters. Subsequent to its publication, many variants were proposed, with hybrid architectures with both features of ViTs and CNNs. ViTs have found application in image recognition, image segmentation, weather prediction, and autonomous driving.

9.

Sharpening: The teacher network's output is sharpened using a softmax function with a lower temperature. This makes the teacher more "confident" in its predictions, forcing the student to learn more meaningful representations to match the teacher's sharpened output. Centering: The teacher network's output is centered by averaging it with its previous outputs. This prevents the teacher from becoming biased towards any particular output value, encouraging the student to learn a more diverse set of features. In January 2024, Meta AI Research released an updated version called DINOv2 with improvements in architecture, loss function, and optimization technique. It was trained on a larger and more diverse dataset. The features learned by DINOv2 were more transferable, meaning it had better performance in downstream tasks. In August 2025, Meta AI Research released DINOv3, an update to DINOv2. It introduced image-text alignment like CLIP. It scaled up the model to 7B parameters and the training dataset to 1.7B images (obtained by diversity-sampling an initial dataset with 17B images). Architecturally, it introduced two improvements: Gram anchoring and axial RoPE (Rotary Positional Embeddings) with jittering. Gram anchoring applies teacher-student self-distillation for the Gram matrix between the feature vectors of the patches of an image. It avoids the previously observed problem of degradation of dense feature maps: While performance on global tasks (like classification) continued to improve, performance on dense tasks (like segmentation) would peak early and then decline, with feature maps becoming noisy. Axial RoPE makes the model more robust to varying image resolutions, scales, and aspect ratios.